**Unit:** 6 Unit 7: Factoring
**Topic:** Difference of Squares

**Name:** ______________
**Date:** __________

## Advanced (Carolina Reaper)

**Q1.** Factor $x^2 - 16$
- (A) $x^2 - 4^2$
- (B) $x^2 + 4^2$
- (C) $x + 4)(x - 4)$
- (D) $x - 4)(x + 4)$
- (E) $2x - 8)(2x + 8)$

**Q2.** Tom is a carpenter who needs to cut a $7^2 - 3^2$ square feet piece of wood.
- (A) 40 square feet
- (B) 49 square feet
- (C) 16 square feet
- (D) 40 square feet
- (E) 28 square feet

**Q3.** Factor $25a^2 - 9b^2$
- (A) $(5a - 3b)(5a - 3b)$
- (B) $(5a + 3b)(5a - 3b)$
- (C) $(5a + 3b)(5a + 3b)$
- (D) $(25a - 9b)(25a + 9b)$
- (E) $9(25 - b^2)$

**Q4.** Solve for $x$ where $x^2 - 9 = 16$
- (A) $x = pm5$
- (B) $x = pm7$
- (C) $x = 5$
- (D) $x = -5$
- (E) $x = 0$

**Q5.** Ashley has $x^2 - 4$ boxes of shoes to pack.
- (A) 2 boxes
- (B) 4 boxes
- (C) x + 2 boxes
- (D) x - 2 boxes
- (E) (x + 2)(x - 2) boxes

**Q6.** Factor $36 - y^2$
- (A) $y^2 - 36$
- (B) $6^2 - y^2$
- (C) $(6 - y)(6 + y)$
- (D) $6y - 36y$
- (E) $6y + 36y$

**Q7.** A bookshelf has a square base with area $49 - x^2$ square feet.
- (A) $7 - x$ feet by $7 - x$ feet
- (B) $7 - x$ feet by $7 + x$ feet
- (C) $7 + x$ feet by $7 - x$ feet
- (D) $7 + x$ feet by $7 + x$ feet
- (E) $7 - x$ feet by $7 - x$ feet

**Q8.** Solve for $x$ where $x^2 - 4x - 21 = 0$ by difference of squares
- (A) $x = pm7$ and $x = pm3$
- (B) $x = 7$ and $x = 3$
- (C) $x = -7$ and $x = -3$
- (D) $x = -7$ and $x = 3$
- (E) $x = 3$ and $x = 7$

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**Open Ended Questions (FRQ)**

**Q9.** If the area of a room is given by $x^2 - 9$, find its dimensions and sketch its graph. What are the limitations of this method?

**Q10.** Compare and contrast the factoring of $x^2 + 9$ with $x^2 - 9$. Provide examples from shopping and explain the limitations of this method.

**Chef's Tips**



- Tip 1: Encourage students to check their answers using a different method.
- Tip 2: Provide opportunities for students to apply difference of squares to problems involving time and money.
- Tip 3: Remind students to always factor out the common term first, if possible.